Project 3 - Basic RMAN

Contents

1.Setting up the environment for the backup
2.Understanding RMAN environment
3.Backing up the cbd
4.Backing up a pdb
5.Point-in-time recovery
6.Errors and troubles
7.Resources

1.Setting up the environment for the backup

There are some prerequisites we have to make sure are enabled:

1. The archive mode is enabled. This is essential for point-in-time recovery. To check this we will use the command ARCHIVE LOG LIST;

```
Modo Log de la Base de Datos           Modo Archive Log
Archivado Automático                   Activado
Destino de Archivado                   USE_DB_RECOVERY_FILE_DEST
Secuencia de log en línea más antigua  88
Siguiente secuencia de log para archiva 90
Secuencia de log actual                90
```

   In our case it is activated. How could we enable it?

```
--Enabling the archive log mode
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
alter database archivelog;
ALTER DATABASE OPEN;
ALTER PLUGGABLE DATABASE ALL OPEN;
```

2. Configure the fast recovery area, the place where we will store the backup and the recovery files. Let's check two parameters which manage it.

```
NAME                       TYPE        VALUE
-------------------------- ----------- -----
db_recovery_file_dest_size big integer 20G
NAME                       TYPE        VALUE
-------------------------- ----------- ------------------------------------------
db_recovery_file_dest      string      \app\sauky\product\23ai\fast_recovery_area
db_recovery_file_dest_size big integer 20G
```

We could change their values by using ALTER SYSTEM SET parameter_name = value;

2.Understanding RMAN environment

First of all, we login to RMAN:

```
C:\Users\sauky>rman target=/

Recovery Manager : Release 23.0.0.0.0 - Production on MiÚ Mar 25 16:10:15 2026
Version 23.6.0.24.10

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conectado a la base de datos de destino: FREE (DBID=1489294439)

RMAN>
```

If we use the command *list backup of database;* we will see the available backup we did before. We will delete them and create new ones.

```
RMAN> delete backup;
delete backup;
canal asignado: ORA_DISK_1
canal ORA_DISK_1: SID=31 tipo de dispositivo=DISK

Lista de Partes de la Copia de Seguridad
Clave BP  Clave BS  N·mero de Parte N·mero de Copia Estado      Tipo de Dispositivo Nombre de Parte
------- ------- --- --- ----------- ----------- ----------
28      28      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\BACKUP
SET\2026_01_13\O1_MF_ANNNN_TAG20260113T143808_NPDLP3V1_.BKP
29      29      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\BACKUP
SET\2026_01_13\O1_MF_NNNDF_TAG20260113T143813_NPDLP5OV_.BKP
30      30      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\20EE36
5E407B4D04895C246AA116763A\BACKUPSET\2026_01_13\O1_MF_NNNDF_TAG20260113T143813_NPDLPO1J_.BKP
31      31      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\A2EDF6
6EA2724525867CAF3717AAE003\BACKUPSET\2026_01_13\O1_MF_NNNDF_TAG20260113T143813_NPDLPW9T_.BKP
32      32      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\4ECEFC
DEC71B4133904887F270B9F56E\BACKUPSET\2026_01_13\O1_MF_NNNDF_TAG20260113T143813_NPDLQCFZ_.BKP
33      33      1   1   AVAILABLE   DISK        C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\BACKUP
SET\2026_01_13\O1_MF_ANNNN_TAG20260113T143858_NPDLQLX3_.BKP
```

Now, If we try to list the backups we won't see anything.

```
RMAN> list backup of database;
list backup of database;
la especificaci¾n no coincide con ninguna copia de seguridad del repositorio
```

We can also list the archivelogs we have.

```
RMAN> list archivelog all;
list archivelog all;
Lista de Copias de Archive Log para la Base de Datos con db_unique_name FREE
=====================================================================

Clave     Thrd Sec.     S Hora Inferior
------- ---- ------- - -------------
14      1    6       A 13/01/26
        Nombre: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\ARCHIVELOG\2026_01_14\O1_MF_1_6_NPHRSMK8_.ARC

15      1    7       A 14/01/26
        Nombre: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\ARCHIVELOG\2026_01_14\O1_MF_1_7_NPHRW41H_.ARC

16      1    8       A 14/01/26
        Nombre: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\ARCHIVELOG\2026_01_14\O1_MF_1_8_NPHSGZ30_.ARC

17      1    9       A 14/01/26
        Nombre: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\ARCHIVELOG\2026_01_16\O1_MF_1_9_NPNZPW3B_.ARC

18      1    10      A 16/01/26
        Nombre: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\ARCHIVELOG\2026_01_17\O1_MF_1_10_NPPLJ8DR_.ARC
```

In order to perform a backup we will use the following command:

To check the backup we will use *list backup of database;*

```
RMAN> list backup of database;
list backup of database;

Lista de Juegos de Copias de Seguridad
==================


Clave BS  Tipo LV Tama±o       Tipo de Dispositivo Tiempo Transcurrido Hora de Finalizaci¼n
------- ---- -- ---------- ----------- ------------ --------------------
52      Full    1.65G      DISK        00:00:10     25/03/26
        Clave BP: 52   Estado: AVAILABLE  Comprimido: NO  Etiqueta: TAG20260325T172522
        Nombre de Parte: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\BACKUPSET\2026_03_25\O1_MF_NNNDF_TAG20260
325T172522_NW833LJH_.BKP
  Lista de Archivos de Datos en el juego de copias de seguridad 52
  File LV Type Ckp SCN    Hora de Punto de Control Abs Fuz SCN Sparse Name
  ---- -- ---- ---------- ------------------------ ----------- ------ ----
  1       Full 34503031   25/03/26                              NO    C:\APP\SAUKY\PRODUCT\23AI\ORADATA\FREE\SYSTEM0
1.DBF
  3       Full 34503031   25/03/26                              NO    C:\APP\SAUKY\PRODUCT\23AI\ORADATA\FREE\SYSAUX0
1.DBF
```

We can see backup sets (each set contains pieces) (BS) from 52 to 57 and all of them belong to the same backup due to all of them sharing the same tag value: TAG20260325T172522. Each backup set belongs to a pdb or the cbd.

By using *list backup of controlfile;* we will see the controlfile information.

```
RMAN> list backup of controlfile;
list backup of controlfile;

Lista de Juegos de Copias de Seguridad
===================


Clave BS  Tipo LV Tamaño       Tipo de Dispositivo Tiempo Transcurrido Hora de Finalizaci¾n
------- ---- -- ---------- ----------- ------------ --------------------
59      Full    18.05M     DISK        00:00:00     25/03/26
        Clave BP: 59   Estado: AVAILABLE  Comprimido: NO  Etiqueta: TAG20260325T172631
        Nombre de Parte: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE\AUTOBACKUP\2026_03_2
5\O1_MF_N_1228843591_NW835QMS_.BKP
  Archivo de Control Incluido: SCN de Punto de Control: 34503088     Hora de Punto de Control:
25/03/26
```

Finally, we can see the spfile backup information.

```
RMAN> list backup of spfile;
list backup of spfile;
la especificaci¾n no coincide con ninguna copia de seguridad del repositorio


RMAN> show controlfile autobackup;
show controlfile autobackup;
Los parßmetros de configuraci¾n de RMAN para la base de datos con db_unique_name FREE son:
CONFIGURE CONTROLFILE AUTOBACKUP ON; # default


RMAN> backup spfile;
backup spfile;
Empezando backup a las 27/03/26
canal asignado: ORA_DISK_1
canal ORA_DISK_1: SID=214 tipo de dispositivo=DISK
RMAN-00571: ===========================================================
RMAN-00569: =============== ERROR MESSAGE STACK FOLLOWS ===============
RMAN-00571: ===========================================================
RMAN-03002: fallo del comando backup en 03/27/2026 12:44:54
RMAN-06062: no se puede realizar una copia de seguridad de SPFILE ya que no se ha iniciado la instancia con SPFILE
```

In this case, we get an error because we started up the database by using a pfile for this reason we can't backup the spfile.

3.Backing up the cbd

We will do some transactions before restoring the database. We will see how this table we had created after doing the backup will be restored to by using the archive redo log.

```
--Before carry out the backup we will execute some transactions
show pdbs;
ALTER SESSION SET CONTAINER=pdb_nebula;
CREATE TABLE NEBULA.CENTROS_MEDICOS( nombre VARCHAR2(50),
                                     ciudad VARCHAR2(50));
DESC NEBULA.CENTROS_MEDICOS;
INSERT INTO NEBULA.CENTROS_MEDICOS(nombre, ciudad)
                            VALUES('Joaquín Sorolla', 'Valencia');
INSERT INTO NEBULA.CENTROS_MEDICOS(nombre, ciudad)
                            VALUES('Hospital de la Princesa', 'Madrid');

SELECT * FROM NEBULA.CENTROS_MEDICOS;
```

Now, get down to business, let's restore the backup. We will use the following script: IMPORTANT NOTE: It is recommended to close all sessions so the backup runs sooner.

```
RUN
{
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
RESTORE DATABASE;--put the database files again in the original location
RECOVER DATABASE;--recover the archive redo log
ALTER DATABASE OPEN;
}
```

Finally, we want to check if all the information was restored so we will do a Select statement to see the information in the CENTROS_MEDICOS table. Thanks to the redo log we didn't lose any information.

4.Backing up a pdb

Firstly, we are going to perform a pdb backup from the database we are interested in. Before that we will check if we have any backup for this pdb:

```
RMAN> list backup of database pdb_nebula;
```

```
Clave BS  Tipo LV Tamaño       Tipo de Dispositivo Tiempo Transcurrido Hora de Finalización
------- ---- -- ---------- ----------- ------------ --------------------
55      Full    727.20M    DISK        00:00:05     25/03/26
        Clave BP: 55   Estado: AVAILABLE  Comprimido: NO  Etiqueta: TAG20260325T172522
        Nombre de Parte: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE
\940EBD869C1F4B9EAEEA149FA35128D8\BACKUPSET\2026_03_25\O1_MF_NNNDF_TAG20260325T172522
_NW835095_.BKP
  Lista de Archivos de Datos en el juego de copias de seguridad 55
  ID de Contenedor: 8, Nombre de PDB: PDB_NEBULA
  File LV Type Ckp SCN    Hora de Punto de Control Abs Fuz SCN Sparse Name
  ---- -- ---- ---------- ------------------------ ----------- ------ ----
  47      Full 34503055   25/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\SYSTEM01.DBF
  48      Full 34503055   25/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\SYSAUX01.DBF
  49      Full 34503055   25/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\UNDOTBS01.DBF
  50      Full 34503055   25/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\USERS01.DBF
```

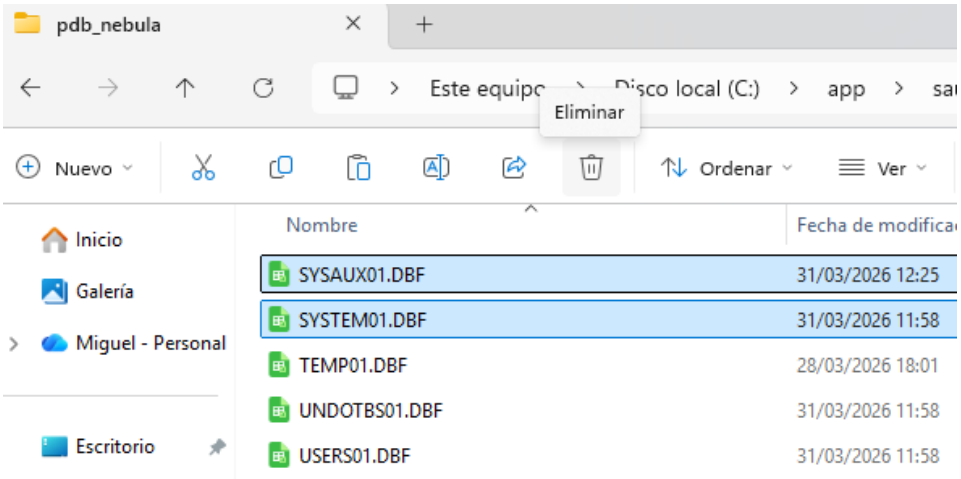
I have a unique backup set (55) for this pdb. To continue, we're going to create a new backup but only for this pdb.

```
backup database pdb_nebula plus archivelog;
```

```
Clave BS  Tipo LV Tamaño       Tipo de Dispositivo Tiempo Transcurrido Hora de Finalización
------- ---- -- ---------- ----------- ------------ --------------------
61      Full    711.05M    DISK        00:00:02     31/03/26
        Clave BP: 61   Estado: AVAILABLE  Comprimido: NO  Etiqueta: TAG20260331T122532
        Nombre de Parte: C:\APP\SAUKY\PRODUCT\23AI\FAST_RECOVERY_AREA\FREE
\940EBD869C1F4B9EAEEA149FA35128D8\BACKUPSET\2026_03_31\O1_MF_NNNDF_TAG20260331T122532
_NWQ88X56_.BKP
  Lista de Archivos de Datos en el juego de copias de seguridad 61
  ID de Contenedor: 8, Nombre de PDB: PDB_NEBULA
  File LV Type Ckp SCN    Hora de Punto de Control Abs Fuz SCN Sparse Name
  ---- -- ---- ---------- ------------------------ ----------- ------ ----
  47      Full 36745265   31/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\SYSTEM01.DBF
  48      Full 36745265   31/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\SYSAUX01.DBF
  49      Full 36745265   31/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\UNDOTBS01.DBF
  50      Full 36745265   31/03/26                              NO    C:\APP\SAUKY\PRODUCT
\23AI\ORADATA\FREE\PDB_NEBULA\USERS01.DBF
```

I can see a new backup set number 61. If we check the TAG we can see they are different because they are from different backups.

Before using the backup we're going to damage our database by deleting some datafiles.



Now, if we try to close the pdb we will get the following error message:

```
SQL> alter pluggable database pdb_nebula close;
alter pluggable database pdb_nebula close
*
ERROR en lÝnea 1:
ORA-01116: error al abrir el archivo de base de datos 48
ORA-01110: archivo de datos 48:
'C:\APP\SAUKY\PRODUCT\23AI\ORADATA\FREE\PDB_NEBULA\SYSAUX01.DBF'
ORA-27041: no se ha podido abrir el archivo
OSD-04002: no se ha podido abrir el archivo
O/S-Error: (OS 2) El sistema no puede encontrar el archivo especificado.
Ayuda: https://docs.oracle.com/error-help/db/ora-01116/
```

So, let's recover our database.

```
RMAN> RUN
2> {
3> SHUTDOWN IMMEDIATE;
4> STARTUP MOUNT;
5> RESTORE PLUGGABLE DATABASE pdb_nebula;
6> RECOVER PLUGGABLE DATABASE pdb_nebula;
7> ALTER PLUGGABLE DATABASE pdb_nebula OPEN;
8> }
```

5.Point-in-time recovery

The point-in-time recovery or PITR is an incomplete recovery which brings our database to a specific point in time. We will recover the pdb_nebula for this exercise.

```
backup database pdb_nebula plus archivelog;
```

Then, let's get a timestamp we will use later to recover the pdb.

```
--Get the timepstamp por the PITR
SELECT TO_CHAR(sysdate, 'DD-MM-YYYY HH24:MI:SS')
FROM DUAL;
--31-03-2026 19:14:36
```

To continue, we will drop a table to restore it after.

```
--I delete the table
DROP TABLE NEBULA.CENTROS_MEDICOS;
COMMIT;
```

Finally, we restore the pdb, recovering the the table CENTROS_MEDICOS:

```
RMAN> RUN
2> {
3> ALTER PLUGGABLE DATABASE pdb_nebula CLOSE;
4> SET UNTIL TIME "TO_DATE('31-03-2026 19:14:36','DD-MM-YYYY HH24:MI:SS')";
5> RESTORE PLUGGABLE DATABASE pdb_nebula;
6> RECOVER PLUGGABLE DATABASE pdb_nebula;
7> ALTER PLUGGABLE DATABASE pdb_nebula OPEN RESETLOGS;
8> }
```

To do a point-in-time recovery of the entire database we will use the following script:

RUN
{
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;
SET UNTIL TIME "TO_DATE('dd-mm-yyyy','DD-MM-YYYY HH24:MI:SS')";
RESTORE DATABASE;
RECOVER DATABASE;
ALTER DATABASE OPEN RESETLOGS;
}

6.Errors and troubles

- The first time I tried to do a backup due to a lack of storage in the recovery area. To solve this issue I looked into the internet for a way to delete archivelog because I had a lot of them. The command I used was: delete archivelog until time ‘SYSDATE-10’; by using this I keep only the archivelog from the last 10 days.

- In order to make it more readable and display the rman information I researched some way to print rman information into a file text. To do this I used: SPOOL LOG TO ‘path\file_name’;

- In the script to recover the pdb_nebula it isn’t necessary to close all, only the pdb we want to recover.

7.Resources

How to delete archivelog properly:
https://dba.stackexchange.com/questions/118686/right-way-to-delete-the-archive-oracle

Create a file text with the RMAN information:
https://docs.oracle.com/cd/B19306_01/backup.102/b14194/rcmsynta058.htm

How to backup the spfile:
https://www.vinchin.com/database-backup/backup-spfile-rman.html